Free Dynamic Pricing with Random Initial Purchase Histories

Toy pricing exercise

1 Timing and initial conditions

Consumers arrive with observed initial state $a \in \{j, k, 0\}$, each with probability $\mu_a = 1/3$ independently of switching type. Initial state 0 has no switching reward. Both periods allow switching. Consumers choose myopically, and period-one outside choices exit permanently.

The firm freely sets transaction prices for both products in both periods. There are no coupons, maintained regular prices, or imposed price bounds. Period-one prices depend on a; period-two prices may depend on the full observed history (a, h), where h is the first purchase. Menus are committed before choices. Type probabilities are known initially, and the objective accounts for endogenous sorting of types into purchase histories. In particular, first purchases can now be informative about type.

2 Consumer and producer problems

For $\ell \in \{j, k\}$ and previous purchase h, deterministic utility is

$$V_{r\ell t}(h) = \beta_\ell + \gamma_r \mathbf{1}\{h \neq 0, \ell \neq h\} + \alpha p_{\ell t}, \qquad s_{r\ell t} = \frac{e^{V_{r\ell t}}}{1 + \sum_m e^{V_{rmt}}}.$$

Outside utility is zero. Period-two utility uses the most recent purchase h, while the firm's prices can retain a as additional information. Define

$$w_r(a, h) = \sum_{\ell \in \{j,k\}} s_{r\ell 2}(a, h)[p_{\ell 2}(a, h) - c_\ell].$$

Value per arriving consumer is

$$\pi_r(P) = \sum_a \mu_a \sum_{h \in \{j,k\}} s_{rh1}(a)[p_{h1}(a) - c_h + \delta w_r(a, h)].$$

The common policy maximizes $q\pi_H(P) + (1 - q)\pi_L(P)$, with $q = \Pr(H)$. Observed-type menus optimize each type separately, using the same cost and initial-state assumptions. There is no additional signal, classifier, targeting cost, or learning intervention.

3 Induction and harvesting

For known type, interior terminal prices satisfy

$$p_{r\ell 2}(h) - c_\ell = -\frac{1}{\alpha s_{r02}(h)}.$$

Product margins are equal at each terminal history. Thus the maintained higher-margin destination of the former coupon exercise does not apply. At an interior first-period optimum,

$$p_{rh1}(a) - c_h + \delta w_r(a,h) = -\frac{1}{\alpha s_{r01}(a)}.$$

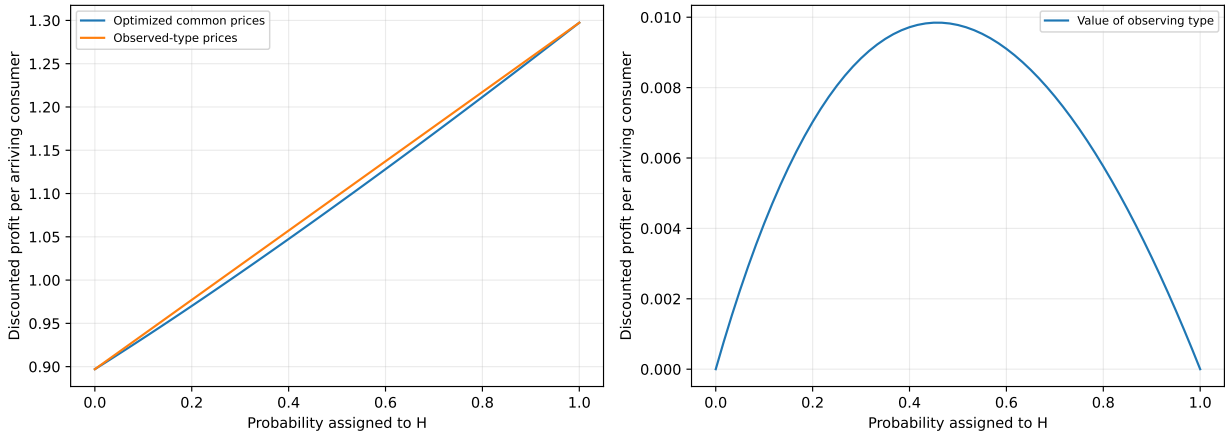
The continuation value acts as a reduction in effective cost. Higher continuation value supports a lower current margin, holding the common effective margin fixed. These known-type equations must not be applied to pooled shares under unobserved heterogeneity.

Report current profit A_r, undiscounted terminal profit B_r per arriving consumer, and total value $A_r + \delta B_r$. Terminal profit includes first-period survival. Profit per survivor divides B_r by its survival probability. For the known-type benchmark, induction sacrifice is the optimized one-period profit minus dynamic current profit; it measures current earnings forgone to support continuation, not a causal coupon effect.

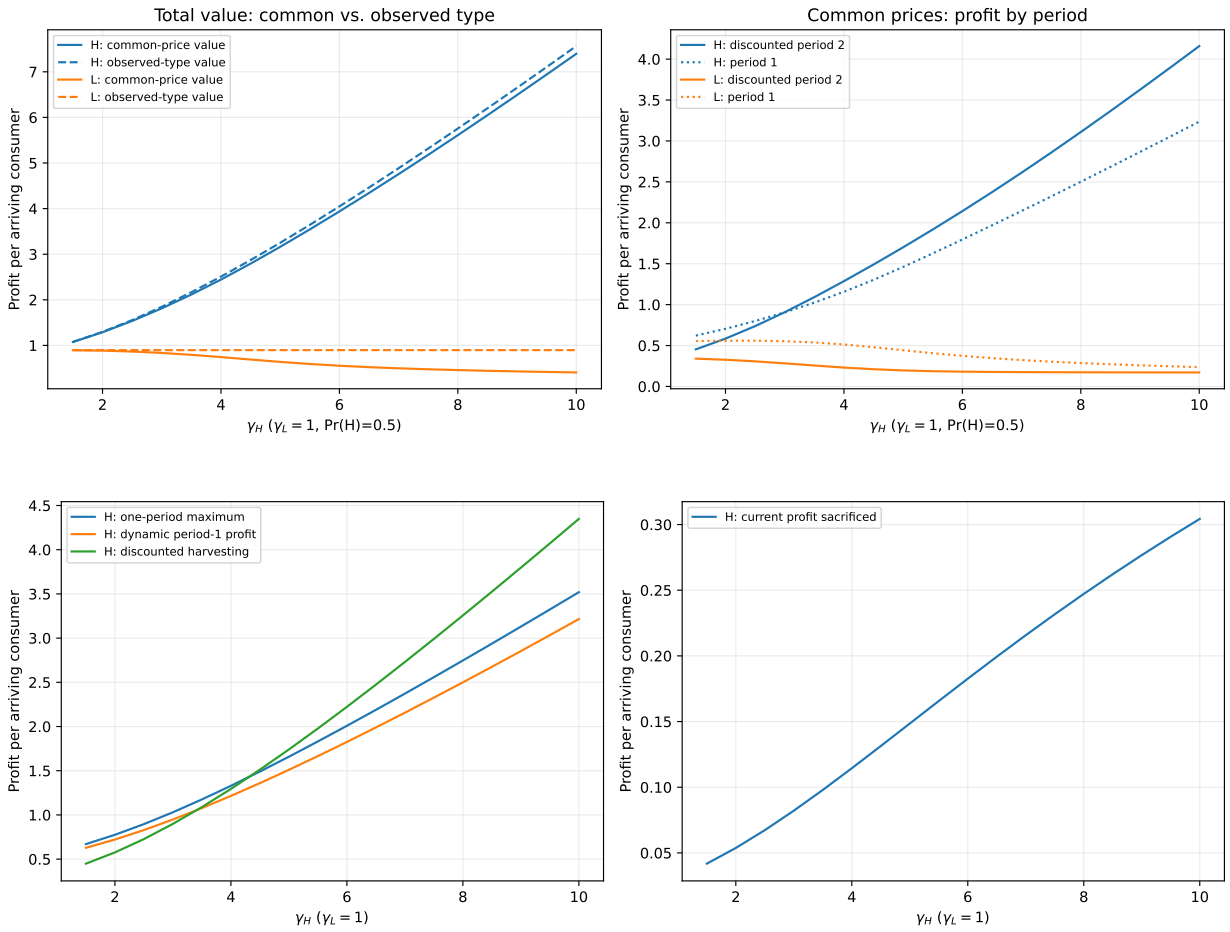
4 Illustration and sensitivity

Parameters are $\beta = (2.2, 2)$, $c = (1.2, 0.8)$, $\alpha = -1.5$, and $\delta = 0.95$. Baseline switching preferences are $(\gamma_H, \gamma_L) = (2, 1)$. Good 2 costs 0.8 versus 1.2 for good 1; both prices remain freely optimized.

At $q = 0.5$, common profit is 1.08744, observed-type profit is 1.09722, and the information gain is 0.00978 per arriving consumer.



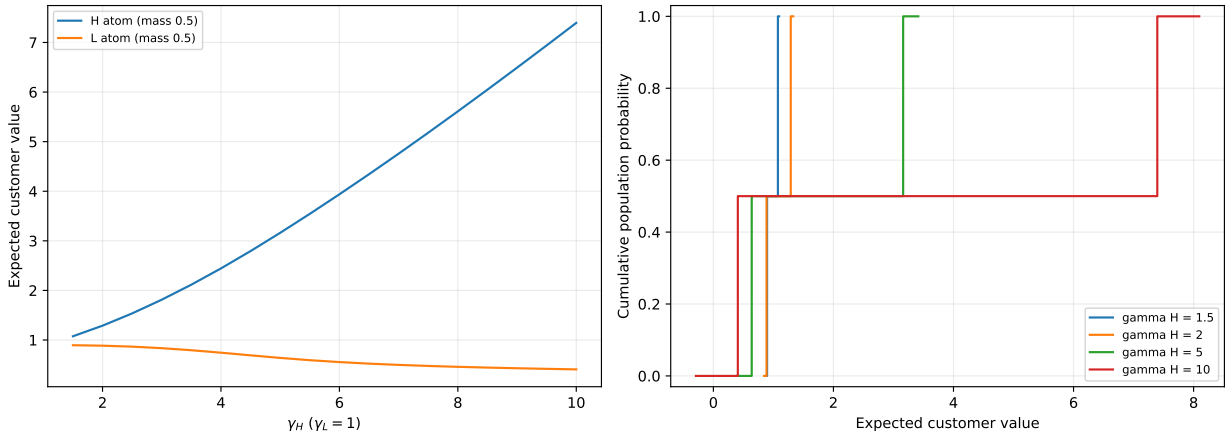
Hold $\gamma_L = 1$ and $q = 0.5$ fixed and vary γ_H from 1.5 to 10 in steps of 0.5, reoptimizing all common prices each time. Known-type values separately optimize each type. A fixed L preference can have changing value under common pricing because the shared menu responds to H.



At each preference pair the distribution of expected customer values is

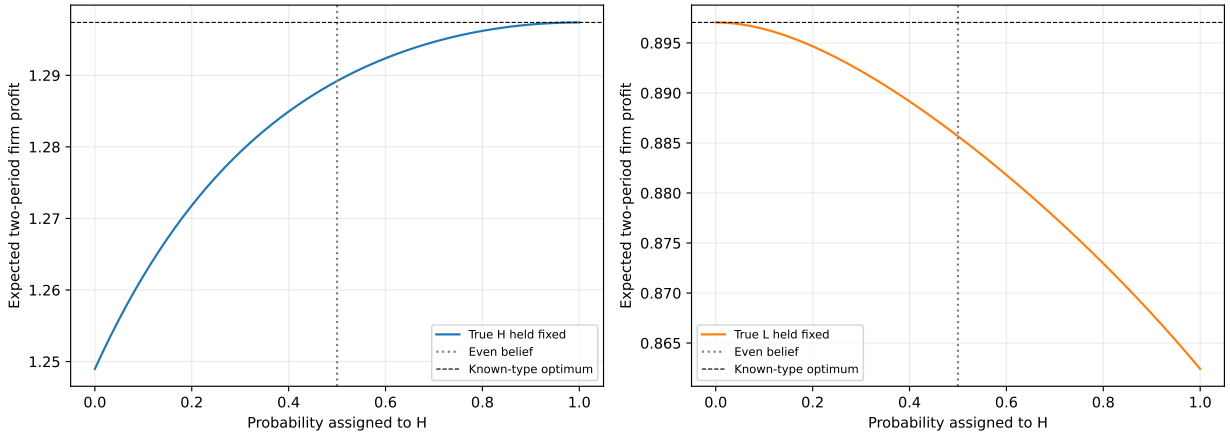
$$F(v) = \tfrac{1}{2}\mathbf{1}\{\pi_H(P^*) \le v\} + \tfrac{1}{2}\mathbf{1}\{\pi_L(P^*) \le v\}.$$

These are two atoms, merging when types coincide. They average over initial states and logit shocks; realized-profit distributions are different. A continuous value density requires a continuous distribution over preferences.



5 Beliefs and mispricing

Hold actual type fixed and evaluate $\pi_r(P^*(q))$. Both panels use the probability assigned to H. Correct certainty is $q = 1$ for H and $q = 0$ for L.



Fixed H, L, and 50/50 menus give affine population-profit lines; their crossings concern a restricted menu choice. Reoptimizing all prices weakly dominates the fixed candidates.

6 Numerical verification and reproducibility

Known-type prices use the backward interior equations. Heterogeneous common menus use three unconstrained BFGS starts per initial state with analytic gradients and a maximum-gradient acceptance threshold of 2×10^{-6}. Multistart optimization provides numerical evidence rather than a global theorem. Tests check finite-difference gradients, initial-state switching, exit accounting, known-type price conditions, information dominance, and comparison with independent derivative-free optimization.

Run the root model, figure builder, and tests as documented in `toy_sol.md`. Decomposition data are in `gamma_H_sweep.csv`; the expected-value atoms are in `value_distribution.csv`. Both are in `results/`. Historical coupon/newborn outputs are not inputs to this exercise.